Rejection Sampling for Medical MCQ Generation and Fine-Tuning TinyLlaMA for Layperson QA

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PowerPoint Slides: Google Drive PDF
Video Presentation: Panopto Video
Code Repository: github.com/chelseanbr/aih-project
Data Source: huggingface.co/datasets/lavita/MedQuAD

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Abstract

Access to accurate, understandable medical information is essential, especially for laypersons without clinical expertise. This project explores how to improve layperson medical question answering (QA) by generating multiple-choice questions (MCQs) from the MedQuAD dataset and fine-tuning TinyLLaMA using a lightweight, parameter-efficient method (LoRA). Through rejection sampling and few-shot Chain-of-Thought prompting, a high-quality MCQ dataset was curated. Fine-tuning led to significant gains over baseline TinyLLaMA and MedAlpaca models, demonstrating the potential of focused domain adaptation even on smaller models.

1 Introduction

Medical QA remains a tough challenge for large language models (LLMs). Medical language is complex, and layperson users require responses that are both medically accurate and easy to understand. Many general-purpose LLMs are not reliably tuned for this domain, leaving significant gaps in accessibility to trusted medical information. To address this, I developed a pipeline to generate high-quality MCQs from the MedQuAD dataset and fine-tuned TinyLLaMA to better handle layperson medical QA. Improving accessibility to medical QA supports broader public health goals by making accurate, understandable answers more available to everyone.

2 Related Work

Ben Abacha and Demner-Fushman [1] proposed an entailment-based medical QA framework, emphasizing the need for relevance and clear justification in answers. Yuan et al. [2] studied how scaling LLM size improves reasoning abilities but pointed out challenges in efficiently adapting large models to specific domains. Zhu et al. [3] demonstrated how synthetic QA data generation, when carefully controlled, can boost medical QA model performance. This project builds on these ideas by combining curated QA generation with efficient fine-tuning for a small, adaptable model.

3 Methodology

The project follows a two-stage pipeline: MCQ generation and TinyLLaMA fine-tuning. The overall process is shown in Figure 1.

3.1 MCQ Generation

I began by sampling 3,000 QA pairs from the MedQuAD dataset. A custom few-shot prompt (Figure 2) with Chain-of-Thought (CoT) examples was designed to guide MCQ generation. Few-shot prompting helped the model produce well-structured questions with reasoning steps.

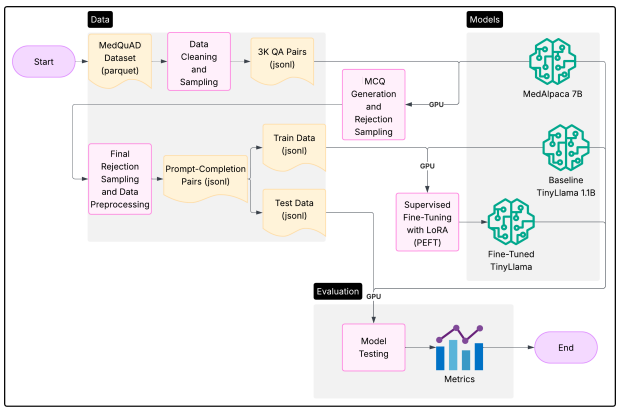

Figure 1: Workflow for MCQ Generation and Fine-Tuning

```
# Few-shot example to encourage structure
few_shot_example = """
Question: What is asthma?
Answer: Asthma is a lung condition that causes airway inflammation and breathing difficulty.
<choices>
A. A skin condition
B. A viral infection
C. A lung disease that causes airway inflammation
D. A type of heart failure
</choices>
<reason>Asthma is a lung condition that causes airway inflammation and breathing difficulty.</reason>
<answer>C</answer>
"""

# Format QA into prompt
def format_prompt(question, answer):
    return f"""
Create layperson multiple-choice answer choices and reasoning for the given medical Q&A pair.
Include 4 total choices: one correct and three plausible, but incorrect. Follow the XML-style structure below:
{few_shot_example}
Now it's your turn:
Question: {question}
Answer: {answer}
"""
```

Figure 2: MCQ Generation Prompt

After generation, rejection sampling was applied to filter outputs for medical correctness, clarity, and formatting. Sampling and generation statistics are shown in Figure 3 and Figure 4.

```
{"question":"What causes Gum (Periodontal) Disease?","answer":"Gum disease is
caused by dental plaque -- a sticky film of bacteria that builds up on teeth.
Regular brushing and flossing help get rid of plaque. But plaque that is not
removed can harden and form tartar that brushing doesn't clean. Only a
professional cleaning by a dentist or dental hygienist can remove tartar."}
```

Figure 3: Sampled Data from MedQuAD

Following manual filtering, 511 MCQs were finalized (Figure 5), with an 80%/20% train/test split.

3.2 Fine-Tuning TinyLLaMA with LoRA

TinyLLaMA was fine-tuned using LoRA (Low-Rank Adaptation), which inserts small trainable matrices into attention layers instead

{"question":"What causes Gum (Periodontal) Disease?","mcq_choices":"A. Genetic mutations\nB. Poor dental hygiene\nC. Injury to the gums\nD. Periodontal abscesses","mcq_reason":"Gum disease is caused by dental plaque — a sticky film of bacteria that builds up on teeth. Regular brushing and flossing help get rid of plaque. But plaque that is not removed can harden and form tartar that brushing doesn't clean. Only a dentist or dental hygienist can remove tartar.","mcq_answer":"B"}

Figure 4: Generated MCQs after Prompting

{"prompt":"You are a helpful and concise medical assistant.\n\nAnswer the following multiple-choice question. First, briefly explain your reasoning using relevant medical knowledge. Then state your final answer letter using exactly one <answer></answer> tag.\n\nInstructions:\n- Use only one <answer> tag (e.g., <answer>B</answer>)\n- Choose the best answer letter\n- Stop immediately after the </answer> tag\n\n<question>\nWhat causes Gum (Periodontal) Disease?\nA. Genetic mutations\nB. Poor dental hygiene\nC. Injury to the gums\nD. Periodontal abscesses\n</question>\n","completion":"Gum disease is caused by dental plaque — a sticky film of bacteria that builds up on teeth. Regular brushing and flossing help get rid of plaque. But plaque that is not removed can harden and form tartar that brushing doesn't clean. Only a dentist or dental hygienist can remove tartar.\n<answer>B</answer>"}

Figure 5: Final Train/Test Split

of updating the full model. This enables efficient adaptation with limited resources.

Fine-tuning parameters:

- LoRA Rank: 8
- LoRA Alpha: 32
- Optimizer: AdamW, learning rate 1e-3

The model was trained using supervised learning to predict the correct MCQ answer label.

4 Results

4.1 Answer Distributions

Figure 6 confirms that the train/test splits were balanced. Figure 7 shows that baseline TinyLLaMA was biased toward a single answer choice. Fine-tuning improved answer distribution (Figure 8), while MedAlpaca (Figure 9) showed partial improvement.

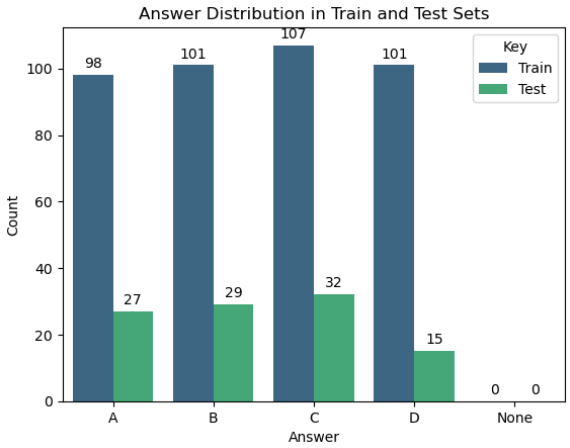

Figure 6: Train/Test Answer Distribution

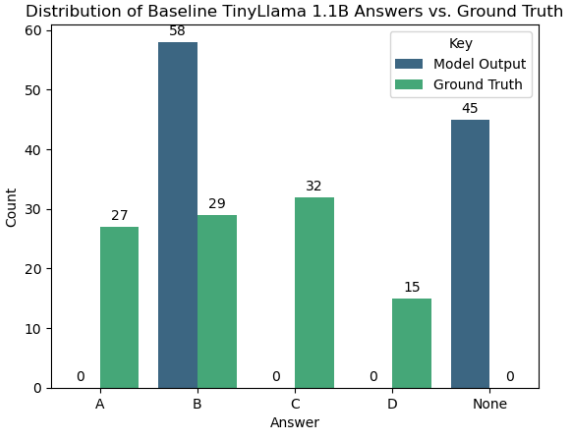

Figure 7: Baseline TinyLLaMA Answer Bias

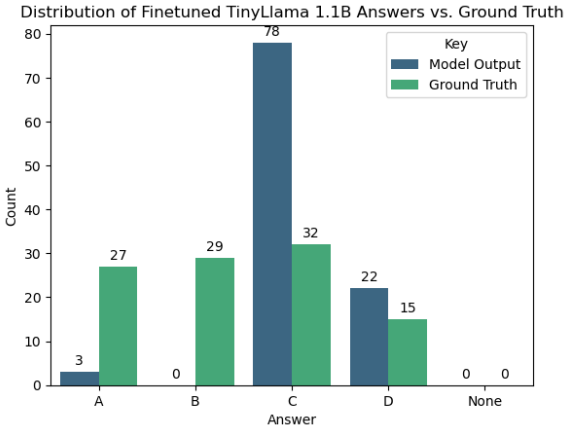

Figure 8: Fine-Tuned TinyLLaMA Answer Distribution

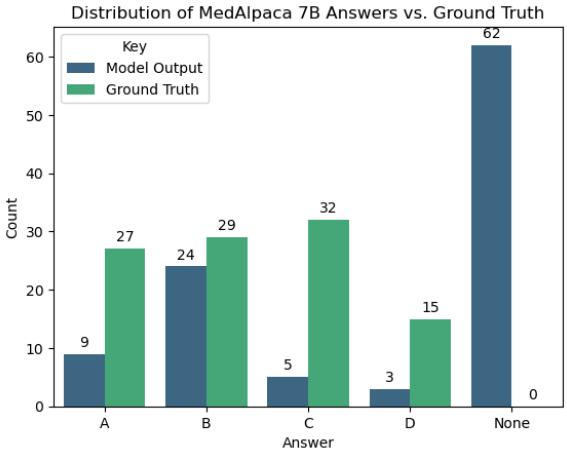

Figure 9: MedAlpaca Answer Distribution

4.2 Accuracy

As shown in Figure 10, fine-tuned TinyLLaMA achieved 33.98% accuracy, improving substantially over the 14.56% baseline and 18.45% MedAlpaca.

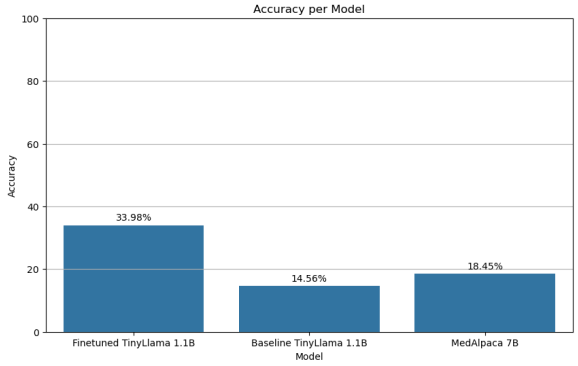

Figure 10: Accuracy Comparison Across Models

4.3 Training Dynamics

Training and validation loss curves (Figure 11) show consistent decline without signs of overfitting, supporting the stability of LoRA fine-tuning even on smaller datasets.

[48/48 06:11, Epoch 3/3]

Epoch	Training Loss	Validation Loss
1	0.846100	0.489880
2	0.370500	0.475589
3	0.363300	0.481885

Figure 11: Training and Validation Loss Curves

5 Conclusion

This project developed a full pipeline for improving layperson medical QA by combining rejection-sampled MCQ generation with efficient LoRA fine-tuning on TinyLLaMA. Results showed significant accuracy gains despite working with limited data and model size.

In future work, I plan to scale fine-tuning to larger models like LLaMA-2 and Mistral, explore open-ended medical QA tasks, apply reinforcement learning (RLHF) for better question generation, and benchmark across more diverse medical QA datasets.

References

[1] Asma Ben Abacha and Dina Demner-Fushman. 2019. A Question-Entailment Approach to Question Answering. *BMC Bioinformatics* 20, 1 (2019). https://bmcbioinformatics.biomedcentral.com/articles/10.1186/s12859-019-3119-4 Article number 511:1–511:23.

[2] Zheng Yuan, Hongyi Yuan, Chengpeng Li, Guanting Dong, Keming Lu, Chuanqi Tan, Chang Zhou, and Jingren Zhou. 2023. Scaling Relationship on Learning Mathematical Reasoning with Large Language Models. *arXiv preprint arXiv:2308.01825* (2023). https://arxiv.org/abs/2308.01825

[3] Jiean Zhu. 2024. Cura-LLaMA: Evaluating Open-Source Large Language Model's Question Answering Capability on Medical Domain. *Applied and Computational Engineering* 90 (2024), 52–60. doi:10.54254/2755-2721/90/20241725